



Can Heparin-Calibrated Anti-Xa Assays Reliably Estimate Direct Oral Factor Xa Inhibitor Levels? Analytical Evaluation in Routine Clinical Practice

Bounaas S^{1,2}, Bouhedadja Z¹, Boulemkahel N², Kehila S^{1,2}, Djerdi R^{1,2}, Lasgaa RM^{1,2}, Kacem M^{1,2}, Zouaghi S^{1,2}

1. Université Ferhat Abbas de Sétif. Faculté de médecine.

2. Laboratoire central.CHU de Sétif.



Introduction

Direct oral anticoagulants (DOACs), particularly the factor Xa inhibitors rivaroxaban and apixaban, are widely used for the prevention and treatment of thromboembolic disorders. In Algeria, routine measurement of DOAC plasma concentrations remains limited by restricted access to specific calibrators and specialized hemostasis laboratories. Consequently, low-molecular-weight heparin (LMWH)-calibrated anti-Xa assays are frequently used as an alternative despite limited local validation. This study evaluated the analytical performance and clinical concordance of a heparin-calibrated anti-Xa assay compared with the reference method using DOAC-specific calibrators.

Methodology

A retrospective analytical study was conducted at the Hemostasis Laboratory of the University Hospital of Sétif between December 2025 and May 2026. Twenty-seven patients receiving rivaroxaban (n = 11) or apixaban (n = 16) were included. Fifty-four paired plasma samples collected at pharmacological peak and trough were analyzed on a STA Compact Max® using two chromogenic anti-Xa assays: a reference method with DOAC-specific calibrators (ng/mL) and a hybrid method calibrated with LMWH (IU/mL). Correlation, linear regression, Cohen's kappa coefficient, and ROC curve analyses were performed.

Results

Overall, the two methods showed a strong correlation ($\rho = 0.802$; $p < 0.001$)

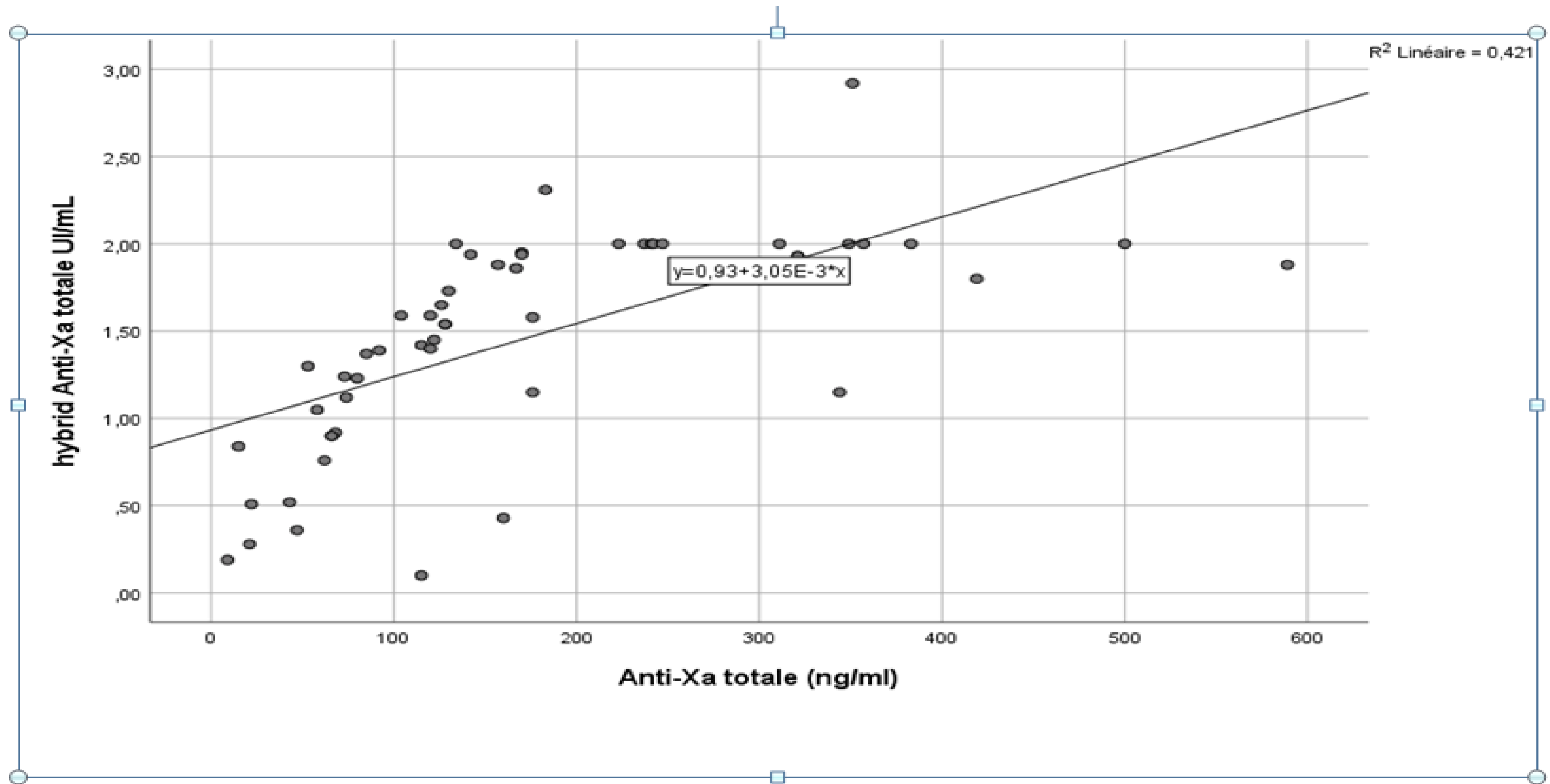


Figure: Corrélation sur la totalité des anti-Xa apixaban et rivaroxaban creux et pic ensemble (n=54) entre les deux méthodes

At trough, significant correlations were observed for rivaroxaban ($\rho = 0.773$; $p = 0.005$) and apixaban ($\rho = 0.612$; $p = 0.012$), whereas no significant correlation was found at peak because of saturation of the LMWH calibrator.

Therapeutic classification concordance was moderate for rivaroxaban ($\kappa = 0.450$) and absent for apixaban ($\kappa = 0.000$)

Tableau 1 : Concordance des classements thérapeutiques du rivaroxaban (n=11) entre la méthode spécifique (étalonnage ISTH 2021) et la méthode hybride HBPM.

		Rivaroxaban — Méthode spécifique ISTH 2021 (ng/mL)			Total
		Infra-thérapeutique < 12	Intra-thérapeutique 12 – 343	Supra-thérapeutique > 343	
Méthode hybride HBPM (UI/mL) creux	< 0,2 (infra)	1	0	0	1
	0,2 – 1,5 (intra)	0	8	2	10
	> 1,5 (supra)	0	0	0	0
Total		1	8	2	11

HBPM – héparine de bas poids moléculaire. Catégories spécifiques ISTH 2021 [1]: infra < 12 ng/mL, intra 12–343 ng/mL, supra > 343 ng/mL. Seuils anti-FXa HBPM creux [2]: < 0,2 / 0,2–1,5 / > 1,5 UI/mL.

Tableau 2 : Concordance des classements thérapeutiques De l'apixaban (n=16) entre la méthode spécifique (étalonnage ISTH 2021) et la méthode hybride HBPM.

		Apixaban — Méthode spécifique ISTH 2021 (ng/mL)			Total
		Infra-thérapeutique <41	Intra-thérapeutique 41-321	Supra-thérapeutique > 321	
Méthode hybride HBPM (UI/mL) creux	< 0,05(infra)	0	0	0	0
	0,05 – 1,6(intra)	0	8	0	8
	> 1,26(supra)	0	8	0	8
Total		0	16	0	16

HBPM – héparine de bas poids moléculaire. Catégories spécifiques selon Gosselin et al. [3]: infra < 41 ng/mL, intra 41–321 ng/mL, supra > 321 ng/mL. Seuils anti-FXa HBPM [4]: < 0,05 / 0,05–1,26 / > 1,26 UI/mL.

ROC analysis demonstrated excellent diagnostic performance for apixaban at trough (AUC = 1.000) and moderate performance for rivaroxaban (AUC = 0.667).

Tableau 3: Performance diagnostique ROC de la méthode hybride anti-Xa (n=27)

AOD – Moment	N	AUC	Sensibilité	Spécificité	VPP	VPN	Seuil optimal
Rivaroxaban – creux	11	0,667	66,7 %	100,0 %	100,0 %	88,9 %	$\geq 1,88$ UI/mL
Rivaroxaban – pic	11	0,357	14,3 %	100,0 %	100,0 %	40,0 %	—
Apixaban – creux	16	1,000	100,0 %	100,0 %	100,0 %	100,0 %	$\geq 2,00$ UI/mL
Apixaban – pic	16	0,625	75,0 %	75,0 %	50,0 %	90,0 %	$\geq 1,93$ UI/mL

Conclusion

The LMWH-calibrated anti-Xa assay may serve as a practical screening tool for estimating residual DOAC concentrations, particularly at trough in emergency settings where specific calibrators are unavailable. However, it cannot replace DOAC-specific calibration for accurate quantification, especially at peak concentrations. Larger multicenter studies are warranted to define standardized decision thresholds and support its implementation in routine laboratory practice.